

HCI History and Opportunities in 2022 - What We Anticipated, What We Did Not

Jonathan Grudin

Microsoft

Redmond, WA, United States

University of Washington

Seattle, WA, United States

jgrudin@microsoft.com

ABSTRACT

Human-computer interaction has entered a third, globally-connected era. The initial focus on making computers usable was followed by efforts to realize visions of the potential, with CHI a key player. Those visions are realized, yielding new opportunities and challenges, many of them unanticipated. HCI draws on computer science, human factors, information systems, and information science. It relies on design and interacts with AI. This course provides an understanding of forces that guide the interaction of related disciplines, constraints imposed by human nature and the trajectories we are following, and opportunities and issues that will engage us in the years ahead.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Usability testing;**

KEYWORDS

Rank, Comparison, Prioritization, Decision-making

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1 BENEFITS OF UNDERSTANDING HCI HISTORY AND ITS RELATIONSHIP TO CHALLENGES TODAY

Participants will learn how and why the disciplines that contributed to HCI differ in their objectives and methods. They will get a deeper sense of how human nature can constrain technology uses, important consequences of computing that were not anticipated, and the opportunities and challenges at this point in our trajectory.

It is said that those who do not learn history are doomed to repeat it. This is not so often the case in our rapidly changing technological world. Fifteen years ago, a new HCI rose from the ashes

of an Internet bubble. Nevertheless, we risk reinventing things that were tried and failed. When past results came from technology limitations, success is quite possible as technology advances. When the constraints were rooted in human cognitive and social behavior, history could repeat itself. When organizational behavior was the key factor, new efforts could succeed as cloud computing and globalization change how we work and live.

Recent years have seen major changes in our interactions with devices and software. Many are positive, but there is growing uneasiness about effects of their uses and about the organizations that control the technologies. As billions of people increase their reliance on technologies that are changing rapidly, unintended consequences can have profound effects that are difficult to anticipate, spread rapidly, and may be difficult to steer. By looking back to find patterns in what was not anticipated before, we may better anticipate and design for the future.

2 INTENDED AUDIENCE

A researcher or developer from any field who has a focus on technology design and use can benefit. In addition to better understanding the temporal dimension, it can be useful to understand what different disciplines engaged in HCI bring, to see which might be relevant to your work and which are not likely to be relevant. Students can see how to position their work, learn from the past, and identify considerations that have emerged and deserve attention. I started as a practitioner and was motivated to explore HCI history to better understand things I learned the hard way. Many of these are still potentially useful, either directly or by analogy. Everyone can profit from understanding tradeoffs that have shaped the HCI trajectory they are now riding.

3 PREREQUISITES AND HANDS-ON ACTIVITY

There are no prerequisites or hands-on work, beyond opportunities for questions and discussion.

4 COURSE STYLE AND CONTENT

This 80-minute hybrid lecture course will invite questions or comments prior to and during the course. My slides rely on timelines, images, and quotations, minimizing bullet points. Previous presentations followed the outline of my 2017 book, *From Tool to Partner: The Evolution of Human-Computer Interaction*. This course will draw from that material, but events over the past five years have not only revealed opened up new opportunities and confronted us with new challenges, they have revealed the critical importance of

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past events that were not evident in 2017. The course starts with the ongoing challenge of HCI arising from rapid changes in technology that can collide with human nature, a familiar observation but one that deserves a careful look. The course then surveys HCI history across fields that have been significantly engaged, but do not always communicate. The history of human-computer interaction falls into three eras: Prior to 1980, when computer processing was very expensive but some envisioned the future potential, from 1980 to 2005 when lower cost led to the rapid expansion of discretionary use and the realization of the early visions, and the fundamentally new era of HCI that presents opportunities and challenges not previously imaginable. Both transitions were disruptive.

Among the disciplinary considerations are the relationship of AI and HCI, and within HCI the balance of quantitative and qualitative approaches. More recent developments include the relationship of interaction design and visual design, and the growing diversity of customer and user goals. A longstanding challenge has been the role of digital technologies in supporting and disrupting communities. A topic moving front and center with surprising ferocity is the proliferation of unintended negative consequences of our globe-spanning work, in the hands of both the well-intentioned and the increasingly powerful bad actors, with many operating along a gray spectrum in between.

The historical review covers the fields of human factors, management information systems, computer science, and information science. It draws from articles and chapters in *Annals of the History of Computing*, *AI Magazine*, *CACM*, *IEEE Computer*, and several

handbooks, an *Interactions* column on HCI history that I wrote and edited for several years, all of which informed by book. And it includes my recent reassessment of that past.

5 INSTRUCTOR BACKGROUND

I have been involved with CHI since it began, contributing to methods and especially to the use of collaboration technologies in organizations. I was Professor of Information and Computer Science at UC Irvine, taught at Aarhus University, Keio University, and the University of Oslo, before joining Microsoft. I am an ACM Fellow and CHI Academy member. My recent work has focused on education, AI technology, and a collaboration with a team that is studying novel crowdsourcing approaches in China.

6 RESOURCES

My web site is jonathangrudin.com. My book *From Tool to Partner: The Evolution of Human-Computer Interaction* is available through Morgan Claypool site license at many universities. For several years I edited a series of essays on HCI history for *Interactions* magazine. My online blog posts for *Interactions* covered a wide range of observations that could be of interest, the most recent being on how lessons learned from early research with streaming media prototypes for higher education could be applied to the remote learning that was forced by COVID, an example of how work conducted before the necessary technology is widely-available can later become relevant.